



Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b) {
5      long long x = *(long long*)a;
6      long long y = *(long long*)b;
7      if (x > y) return 1;
8      if (x < y) return -1;
9      return 0;
10 }
11
12 int main() {
13     int n;
14     scanf("%d", &n);
15
16     long long arr[n];
17     for (int i = 0; i < n; i++) {
18         scanf("%lld", &arr[i]);
19     }
20
21     qsort(arr, n, sizeof(long long), compare);
22
23     long long max_sum = 0;
24     for (int i = 0; i < n; i++) {
25         max_sum += arr[i] * i;
26     }
27
28     printf("%lld\n", max_sum);
29     return 0;
30 }
```

Check

	Input	Expected	Got	
✓	5	40	40	✓
	2			
	5			
	3			
	4			
	0			

	Input	Expected	Got	
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

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